SOLUTION DOCUMENT

Rapid Design Engineering

Solution code: SOL-001
Solution title: Process Optimization
Client: Example SME S.A.
Date: 2026-10-21
Version: 1.0

The data, values and results in this sample are illustrative.

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Abbreviated sample · Illustrative content · Not a real case

A real document is 18–25 pages.

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1 Executive summary

Solution summary

Problem in 3 lines: The production line loses time to bottlenecks and rework between stations.

Solution in 3 lines: Process redesign and station balancing with work standards.

Expected benefit: Estimated 20% reduction in cycle times.

Resources needed to implement: Two weeks of assessment and one production owner.

2 Problem identified

2.1 Problem description

Description: The current process has waiting time between stations and rework.

2.2 Root cause

Root cause (not the symptom): Lack of line balancing and work standards.

2.3 Supporting data

Verifiable sources: Production records from the last three months.

2.4 Consequences of not solving it

Consequences: Higher unit cost and delayed deliveries.

3 Alternatives evaluated

Evaluation criteria (cost, time, complexity, risk, standards): Cost, time, complexity, risk and applicable standards.

Alternative	Cost	Time	Risk	Reason for rejection
Alternative A	USD 10,000	8 weeks	Low	Requires initial investment.
Alternative B	USD 6,000	12 weeks	Medium	Longer implementation time.

4 Recommended solution

4.1 Technical description

Description: Reorganize the flow, balance the stations and define work standards to reduce waiting and rework.

4.2 Design and calculation criteria

Parameters and applicable standards: Line balancing, time analysis and capacity definition.

4.3 Implementation conditions

What is needed / what to avoid: Do not stop production during the assessment and ensure staff availability.

4.4 Diagrams and schematics

Insert diagrams, tables or schematics if applicable: Not applicable.

5 Expected result

Measurable benefit (savings, time reduction, efficiency improvement): 20% reduction in cycle times.

Assumptions used in the calculation: Stable demand and staff availability.

6 What this solution does NOT include

Deliverable limits

- Physical prototype.
- Detailed engineering development.
- Project execution.
- Construction management.
- Guarantee of execution results (depends on the implementer).

7 Terms of use and license

License type (use, not sale): License of use.

Prohibition of reproduction and distribution: Resale of the document is not permitted.

Solution validity (if applicable): 12 months.

8 Annexes

8.1 Sources consulted

Sources: Client's internal records.

8.2 Standards cited

Standards: Not applicable.

8.3 Technical glossary

Glossary (if the client is not an engineer): Cycle time: total time for one unit.